

Hai Whakataki | Course Outline

ENGN 676 Agricultural Engineering

Semester 1, 2025

Haere ki Akoraka mō ētahi whakamahuki kia whai tautoko
For information on Lincoln University support and services, please refer to the links on
My Akoraka | Learn.

Examiner / Lecturer	< Majeed Safa R014 Ross Building Email: Majeed.Safa@lincoln.ac.nz Phone: 03 4230263 Contact: The best way to contact me is via email. My usual response time is less than 24 hours during the working week. Book an appointment or come along to my student drop-in time on Thursdays From 9:00 to 11:00. You can meet me online (see the Akoraka Learn course page for the link) or in person in my office.		
Activity	Day	Time	Room
Lecture	Monday	10:00-10:50	D6
Lecture	Thursday	11:00-11:50	D6
Lecture	Friday	9:00-9:50	D6
Lab	Tuesday	11:00-1:00	Structures Lab
You are advised to familiarise yourself with the <i>Code of Conduct for Field Trips</i> on Akoraka Learn			
Teaching on Field Trip Days Face-to-face activities and office hours <i>will not</i> be held on field trips days.			
Course Prescription	Advanced development and management of agricultural and horticultural production, regarding land and water resources, machinery, and infrastructure		
Recommended Preparation	Basic knowledge about New Zealand agriculture and recent agricultural technologies		

Kia whai hua | Course Aims and Learning Outcomes

Aims

The main aim of this course is to develop analytical and design skills in the engineering aspects of agriculture and horticulture, to evaluate and develop technologies relevant to precision agriculture

Learning outcomes

After successfully completing this course, you will be able to:

LO1. Evaluate the key principles of farm operations and the main technological issues in NZ agriculture.

LO2. Evaluate the key principles of surface groundwater resources, water supply and irrigation systems, including the operation of centre pivot and trickle irrigation systems

LO3. Compare and contrast the key principles of new technologies in agriculture including precision agriculture, robotics, and online monitoring

LO4. Develop and apply design philosophy and the design process in agricultural engineering

Te whakahoki kōrero | Course Improvements

The following table outlines the summarised feedback received from previous students (via course evaluations, student rep feedback and other feedback opportunities) and other sources, and how this feedback has been incorporated into the course.

Summary Feedback		Course Changes
Students:	Students expect more practical materials and quest lecturer	So I invited more guest lecturer this year, also I arranged drawing session

Whakatakotoraka | Course Content

The following table gives an indication of the timing of the content for this course. It may be necessary to make adjustments to the timetable.

Week – commencing	Modules / Topics	Online Tutorials & Assignments
1 17 February	Course Introduction Project Discussion	
2 24 February (Field Trip Day: Friday)	Introduction to Design- Design Philosophy	Drone Training
3 3 March	Design Process Advanced Ergonomics in Agriculture	- Guest Lecturer Discussion - Perusall Assignment
4 10 March (Field Trip Day: Thursday)	Physical and Mechanical properties of agricultural production	
5 17 March	Irrigation Systems and Water management	- Guest Lecturer Discussion - Perusall Assignment -Online Quiz
6 24 March (Field Trip Day: Monday)	Tractor and power systems	- Guest Lecturer Discussion -Online Quiz - Perusall Assignment
7 31 March	Drawing 2D and 3D Drawing	
Mon 7 April – Fri 25 April (Public Holiday: Friday 18 April, Monday 21 April, Thursday 25 April) (Uni Holiday: Tuesday 22 April)	Mid Semester Break – includes Easter and ANZAC Day	
8 28 April (Field Trip Day: Wednesday)	Drawing 2D and 3D Drawing	-Drawing Assignment
9 5 May (Graduation: Friday)	Machinery Management and Efficiency on Farms	
10 12 May (Field Trip Day: Tuesday)	Artificial Intelligence & Robotics-	-Online Quiz - Perusall Assignment - Guest Lecturer Discussion
11 19 May	Pest Control Technologies	-Online Quiz - Perusall Assignment
12 26 May	Project Presentation	

Some dates and activities would change

Whakariteka | Learning and Teaching Arrangements

Te Whakatinanataka o te Ako | Learning and Teaching Approach

The learning and teaching approach is based on a combination of face-to-face lectures, interactive lecture/tutorials and on-line resources from the Akoraka | Learn course page. Students are strongly advised to make full use of all available learning opportunities.

The teaching approach taken in this course is to introduce engineering concepts that are then applied to real world problems. The student will have the opportunity to demonstrate their understanding by completing the extra examples supplied at the end of each Module. Another component of this course is to have the students involved in designing an innovative product that will benefit the agricultural community. The course has been designed based on student centered technique and this is supported by weekly group discussions, group interaction activities, individual and group projects and presentations, an individual final project.

Akoraka ā ipuraki | Online Learning Activities

Self-study material, review material, other relevant course material, and assessment activities will be made available on the course Akoraka | Learn page.

You are expected to be able to use MS Teams and Lincoln's Akoraka | Learn system, and have the [basic skills and technology requirements](#) needed to study through the online medium.

All classes will be recorded and these recordings will be available on the Akoraka | Learn course page. The quality of recordings cannot be guaranteed. Guest lecturers may decline to be recorded.

The course may involve online interactive resources, online readings, accessing other online sites, videos, video conferencing, online quizzes, forum posts and submitting or uploading documents/media.

Kōrerorero | Communication

Akoraka | Learn will be used as a means of communication with the class and you are advised to check the course page, the My Akoraka | Learn dashboard, and your "@lincolnuni.ac.nz" email regularly to ensure you receive all University communications. You should also use your "@lincolnuni.ac.nz" email address to communicate with examiners and lecturers.

Kā kupu āwhina | Lecture Notes

Lecture notes will be posted on Akoraka | Learn. It is important to note that the images shown in lectures may not all be available in the pdf version of the notes, as copyright regulations may prevent this. All readings will be placed on the Akoraka | Learn course page.

Poipoia te Ako | Learning Advice

A wide range of learning advice options are available for you to access on- and off-campus. The Learning Advisors in Learning, Teaching and Library (LTL) provide drop-ins, individual appointments, workshops, and resources to support you to further develop your academic writing, referencing, study, and mathematics and statistics skills.

Kā aromatawai | Assessments

The schedule of assessments and their contribution to the overall mark for the course is as follows:

Assessment	Weighting	Due date	Learning outcomes covered
Design project	50%		
- Prototype	10%	20 May	All

- Report	20%	26 May	
- Presentation	20%	27 May	
-			
Perusal Assignments	20%	TBA	All
Online tutorials and Assignments	20%	TBA	LO1-LO3
Guest Lecturer Assignments	10%	TBA	LO1-LO4

Kā whakamahuki aromatawai | Assessment Details

Note: All electronically-submitted assessments will be either completed or uploaded via the Akoraka | Learn course page.

Rīpoata | Report

Kaupapa Tuhika | Assignment / Project / all other written internal assessment

The Final design project is to be submitted by 5 pm on 26 May in dropbox on Learn website

Kauhau | Oral Presentation

A 7–10 minute presentation is expected in the final week to showcase the results of the design project, scheduled for 27 May.

Whakamātau mutuka | Final Examination

There is no final examination in this course.

Kā whakapāha | Aegrotats

Aegrotat applications may be made when due to a *serious event*: (1) a student has missed an assessment; (2) a student has completed an assessment but believe they have been impaired by one of the above circumstances; or (3) a student is seeking an extension to the due date for an assessment. *Serious events* means illness (including COVID-19) or injury; bereavement, trauma, critical circumstances, disasters; and misadventure. To apply for an Aegrotat, your assessment needs to be worth 10% or more of your final course grade. Apply through the online portal on Akoraka | Learn which can be accessed from the link within the Exams & Assessments section on the right of your Akoraka | Learn dashboard. If your Aegrotat application is deemed justified (i.e., approved) by the Examinations Office, the Examiner will decide what the outcome will be - this may include approving an extension, offering an alternative assessment or deriving a grade. For more information on Aegrotats refer to the online portal. For assessments worth less than 10%, you should normally approach the Examiner or their nominee (such as a lecturer or tutor) directly, in person or in writing.

Te whakaroaka i te tuku | Extensions and Late Submission of Assessment

For items of assessment worth less than 10%, you should approach the Examiner or their nominee (such as a lecturer or tutor) directly, in person or in writing. For items of assessment worth 10% or more of your final course grade, if there is a serious event that contributes to you seeking an extension or submitting a piece of assessment late, you should normally approach the Examiner (for extensions of 3 days or fewer) or apply for an Aegrotat (for extensions of more than three days). Also refer to the section on Aegrotats above. The Examiner will decide the outcome of your request - this may include approving an extension or removing any late penalties. If denied, items of assessment (other than tests) that are submitted after the due date and time will have a maximum late penalty applied as noted below. University regulations apply where a final examination is scheduled.

Less than 24 hours late (i.e., 1 day)	10% late penalty
Between 24 and 48 hours late (i.e., 2 days)	20% late penalty
Between 48 and 72 hours late (i.e., 3 days)	30% late penalty
Between 72 and 96 hours late (i.e., 4 days)	40% late penalty
Between 96 and 120 hours late (i.e., 5 days)	50% late penalty
Over 120 hours late (i.e., 5 days)	100% late penalty

Whakature I te akoraka | Mandatory Course Requirements

The following course components are mandatory: Design project .

Failure to submit design project and presentation, without the prior approval of the Examiner, may result in you not being eligible to achieve a passing grade in this course.

If you are unable to complete a mandatory course requirement, please contact the Examiner in advance and if applicable, apply for an Aegrotat.

Te Tū Whakahīhi | Academic Integrity and Behaviour

All forms of academic dishonesty are unacceptable to Lincoln University. This includes, but is not limited to cheating, collusion, double-dipping, fabrication of data, ghost writing, plagiarism, unauthorised use of Artificial Intelligence (AI) to generate content, and possession of unauthorised notes, cell phones or other devices during a test or examination, or access to unauthorised software during a test or examination. Incidents of academic dishonesty may be deemed to be a breach of discipline and may be reported to the Proctor. You will be subject to the terms of the Student Discipline Regulations.

Upon enrolment, you undertake to comply with Lincoln University regulations and policies. Please make yourself familiar with these - including the Lincoln University Student Code of Conduct and the Academic Integrity Policy and Procedure - which can be found on the right-hand side of your Akoraka | Learn page under University Policies and Procedures.

Kā mahi o te tauira | Student Workload

At a minimum, 200 hours for 20-credit courses represents the amount of time that an average or B grade student might be expected to spend to receive a passing grade. The total student workload for a course is not spread evenly from week to week and you are expected to proactively manage their workload throughout the semester. Achievement in a course is based on your performance, not on the time committed to studying the course. No matter how many hours you put into this course, you are not guaranteed a pass. The following time-use guidelines are provided as an example of how the 150 may be allocated in this course.

Indicative Student Workload

Contact Hours (including recorded/online classes)	Total hours (over semester)
Lectures	30
Workshops	20
Tutorials	
Non-contact Hours	
Online Activities	40
Project	110
Work Experience Hours	
Total Student Workload	200

Te whakahoki kōrero | Feedback Opportunities

There will be an opportunity to formally evaluate the course at the end of the semester, however feedback is welcomed and appreciated throughout the semester and may be provided in any format, e.g. in person, with a support person, through a student rep, via a note, or email. You will be able to complete mid-semester feedback in week 6 and this information will be considered for the second half of the semester. Constructive feedback is welcomed and appreciated throughout the semester to allow the Examiner to improve the course and their lecturing style.